

- (GA1) (G.Baumslag) Does the free  $Q$ -group  $F^Q$  act freely on some  $\Lambda$ -tree?
- (GA2) (G.Baumslag, A.Miasnikov, V.Remeslennikov) Is a finitely generated group acting freely on a  $\Lambda$ -tree, automatic (biautomatic)?
- (GA3) (A.Miasnikov) Let  $G$  be a group acting freely on a  $\Lambda$ -tree. Given any finite set of non-trivial elements of  $G \ast G$ , is there a homomorphism of  $G \ast G$  into  $G$  such that the images of the given elements are also non-trivial?
- (GA4) (O.Kharlampovich, A.Miasnikov, V.Remeslennikov) Is the elementary theory of the class of all groups acting freely on a  $\Lambda$ -tree, decidable?
- (GA5) (S.Sidki) Let  $G$  be a group such that it contains a subgroup  $H$  of index 2 and  $H$  admits a homomorphism  $f: H \rightarrow G$ ; we call such a map a 1/2-endomorphism of  $G$ . Then this data can be used to construct a (state-closed) representation  $r$  of  $G$  into the automorphism group of the binary tree where the  $\text{kernel}(r) = \langle K \mid K \text{ subgroup of } H \text{ which is both normal in } G \text{ and } f(K) \text{ is contained in } K \rangle$ . We call  $\text{kernel}(r)$  the  $f$ -core( $H$ ) and say that  $f$  is simple provided  $f$ -core( $H$ ) is the trivial group.
- (a) Which groups admit a simple 1/2-endomorphism?
- (b) If  $G$  admits a simple 1/2-endomorphism, can  $G$  be a free abelian group of infinite rank?
- (c) If  $G$  admits a simple 1/2-endomorphism, can  $G$  be a free group of rank  $k > 1$ ? [Background](#)